

CLICK PLC Ethernet Network Configuration

Configuring an isolated Windows-to-PLC network through a USB Ethernet adapter and unmanaged switch

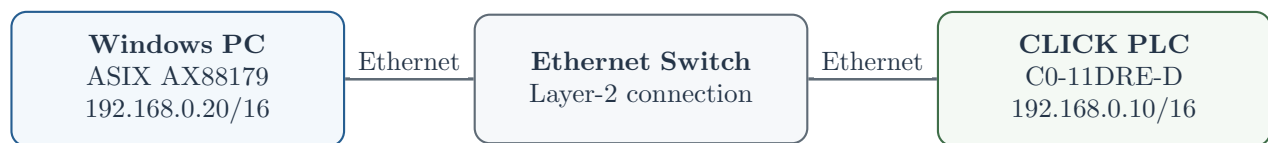
Documented configuration: Windows PowerShell 7.6.4, ASIX AX88179 USB Ethernet adapter, CLICK C0-11DRE-D PLC, and CLICK Programming Software Version 3.92.

Core idea

A Windows computer and CLICK PLC can communicate directly through an Ethernet switch without a router or Internet connection. The two devices must have unique IPv4 addresses on the same subnet. CLICK's LAN discovery can locate the PLC before ordinary IP communication is configured; Windows can then be assigned a compatible static address.

1 Physical Network and Final Address Plan

The test network consists of a Windows computer connected through a USB-C Ethernet adapter to an Ethernet switch. The CLICK PLC is connected to the same switch.



Device	IPv4 Address	Subnet Mask	Notes
Windows USB Ethernet	192.168.0.20	255.255.0.0	Static address; no gateway required
CLICK PLC	192.168.0.10	255.255.0.0	Address discovered with CLICK software
Windows Wi-Fi	192.168.1.147	---	Separate normal LAN/Internet connection

The Ethernet adapter used during configuration had interface index 36 and MAC address 6C-6E-07-1D-63-84. The PLC was discovered with MAC address 00-D0-7C-1B-0A-24.

2 Identify the Windows Ethernet Adapter

Open PowerShell and list the available network interfaces.

Step 1 – List network adapters

```
Get-NetAdapter
```

The USB Ethernet adapter appeared as:

```
Name: Ethernet
InterfaceDescription: ASIX AX88179 USB 3.0 to Gigabit Ethernet Adapter
```

```
ifIndex: 36  
Status: Up  
LinkSpeed: 100 Mbps
```

The Up status confirms that Windows detects a physical Ethernet link through the adapter and switch.

Interface index

The interface index is a convenient way to target one Windows network adapter from PowerShell. In this setup, `-InterfaceIndex 36` selects the ASIX USB Ethernet adapter and avoids accidentally changing the Wi-Fi interface.

3 Inspect the Initial IPv4 Configuration

The initial IP configuration was inspected with:

Step 2 – Inspect the selected interface

```
Get-NetIPConfiguration -InterfaceIndex 36  
Get-NetIPAddress -InterfaceIndex 36 -AddressFamily IPv4
```

Before configuration, Windows reported:

```
IPv4Address: 169.254.227.254  
PrefixLength: 16
```

The address 169.254.227.254/16 is an automatically assigned IPv4 link-local address. It indicated that the Ethernet segment did not provide Windows with an address from a DHCP server.

What the switch does not do

An ordinary unmanaged Ethernet switch does not normally assign IP addresses. It forwards Ethernet frames between devices. Without a DHCP server, Windows can fall back to a 169.254.x.x link-local address while the PLC retains a separately configured address.

4 Check the Initial Neighbor Table

Before the PLC address was known, the Windows neighbor table was inspected.

Step 3 – Inspect learned IPv4 neighbors

```
Get-NetNeighbor -InterfaceIndex 36 -AddressFamily IPv4  
arp -a
```

Only broadcast and multicast entries were initially present on interface 36. There was no ordinary unicast PLC entry. This meant Windows had not yet learned an IP-to-MAC mapping for the PLC.

5 Discover the CLICK PLC

CLICK Programming Software Version 3.92 was used to discover the PLC on the local Ethernet segment.

5.1 Open the connection dialog

From the CLICK Programming Software main window, open the **PLC** connection function and select Ethernet as the port type. In the **Connect to CLICK PLC** dialog:

1. Set **Port Type** to **Ethernet**.
2. Select the **ASIX AX88179 USB 3.0 to Gigabit Ethernet Adapter**.
3. Select **In the same LAN (Scan all CLICK PLCs in the LAN automatically.)**.
4. Allow the software to scan the local Ethernet segment.

The scan discovered the following PLC:

Property	Discovered value
IP address	192.168.0.10
Subnet mask	255.255.0.0
Part number	C0-11DRE-D
Firmware	Ver2.60
Mode	STOP
Status	GOOD
Connection	Available
MAC address	00-D0-7C-1B-0A-24

Why discovery worked

CLICK's local-LAN discovery was able to identify the PLC even though the Windows adapter was still using 169.254.227.254. After discovery supplied the PLC's actual IPv4 address and subnet mask, Windows could be configured for ordinary IPv4 communication on the same subnet.

6 Assign a Compatible Static Address to Windows

The PLC already had a usable address, 192.168.0.10/16, so the PLC configuration was left unchanged. Instead, the Windows Ethernet adapter was assigned 192.168.0.20/16.

PowerShell must be opened with **Run as administrator**; otherwise `New-NetIPAddress` can return `Access is denied`.

Step 4 – Configure the Windows Ethernet address

```
New-NetIPAddress '
  -InterfaceIndex 36 '
  -IPAddress 192.168.0.20 '
  -PrefixLength 16
```

A prefix length of /16 corresponds to the subnet mask:

255.255.0.0

No default gateway or DNS server is required for this isolated PLC Ethernet segment.

Keep the PLC network isolated from Internet routing

Do not add a default gateway to this Ethernet adapter unless the network design specifically requires one. Leaving the gateway blank allows the PC's normal Wi-Fi connection to continue handling Internet traffic while the USB Ethernet interface is used for the PLC subnet.

7 Verify the Windows Address

After assigning the address, verify the interface:

Step 5 – Confirm the static IPv4 address

```
Get-NetIPAddress -InterfaceIndex 36 -AddressFamily IPv4
```

The final relevant result was:

```
IPAddress: 192.168.0.20
InterfaceIndex: 36
InterfaceAlias: Ethernet
AddressFamily: IPv4
PrefixLength: 16
PrefixOrigin: Manual
SuffixOrigin: Manual
AddressState: Preferred
```

The **Preferred** state indicates that Windows considers the address usable.

8 Test Communication with the PLC

The simplest end-to-end IPv4 test is ICMP echo using `ping`.

Step 6 – Ping the CLICK PLC

```
ping 192.168.0.10
```

The test produced four successful replies:

```
Reply from 192.168.0.10: bytes=32 time=3ms TTL=128
Reply from 192.168.0.10: bytes=32 time=2ms TTL=128
Reply from 192.168.0.10: bytes=32 time=2ms TTL=128
Reply from 192.168.0.10: bytes=32 time=2ms TTL=128

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

This confirms ordinary IPv4 connectivity from the Windows PC, through the Ethernet adapter and switch, to the CLICK PLC.

9 Verify the PLC in the Neighbor Table

After communicating with the PLC, inspect the Windows neighbor table again:

Step 7 – Confirm the IP-to-MAC mapping

```
Get-NetNeighbor -InterfaceIndex 36 -AddressFamily IPv4
```

The PLC appeared as:

```
IPAddress: 192.168.0.10
LinkLayerAddress: 00-D0-7C-1B-0A-24
State: Stale
```

The **Stale** state is normal. Windows has a cached neighbor mapping but has not recently reconfirmed it. The entry can be refreshed automatically when more traffic is sent.

9.1 ARP interpretation

Because both addresses are on the same 192.168.0.0/16 subnet, Windows does not need a router to reach the PLC. It resolves the PLC's Ethernet MAC address using the local neighbor/ARP mechanism and sends Ethernet frames directly through the switch.

The useful mapping is therefore:

192.168.0.10	↔	00-D0-7C-1B-0A-24
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10 Final Working Configuration

Windows Ethernet adapter	ASIX AX88179 USB 3.0 to Gigabit Ethernet Adapter
Windows interface index	36
Windows Ethernet MAC	6C-6E-07-1D-63-84
Windows IPv4	192.168.0.20/16
PLC model	C0-11DRE-D
PLC firmware	Ver2.60
PLC IPv4	192.168.0.10/16
PLC MAC	00-D0-7C-1B-0A-24
Subnet mask	255.255.0.0
PLC network gateway	Not required for this isolated network
Connectivity test	4/4 ping replies, 0% packet loss

At this point, return to the CLICK Programming Software connection dialog, select the discovered PLC at 192.168.0.10, and use **Connect**. The PC and PLC now share the same IPv4 subnet and can communicate normally.

11 Useful PowerShell Reference

Inspect the physical adapters

```
Get-NetAdapter
```

Inspect the PLC Ethernet interface

```
Get-NetIPConfiguration -InterfaceIndex 36  
Get-NetIPAddress -InterfaceIndex 36 -AddressFamily IPv4
```

Inspect local IPv4 neighbors

```
Get-NetNeighbor -InterfaceIndex 36 -AddressFamily IPv4
```

Test basic PLC reachability

```
ping 192.168.0.10
```

Test a specific TCP service when its port is known

```
Test-NetConnection 192.168.0.10 -Port <port>
```

12 Configuration Sequence Summary

1. Connect the Windows USB Ethernet adapter and CLICK PLC to the Ethernet switch.
2. Use `Get-NetAdapter` to identify the USB Ethernet interface and its interface index.
3. Inspect its current IPv4 address with `Get-NetIPConfiguration` and `Get-NetIPAddress`.
4. Open CLICK Programming Software and scan for CLICK PLCs on the same LAN using the ASIX Ethernet adapter.
5. Record the PLC's discovered IP address, subnet mask, model, firmware, and MAC address.
6. Open PowerShell as Administrator.
7. Assign the Windows Ethernet adapter an unused static address on the PLC's subnet.
8. Verify the Windows address reaches the **Preferred** state.
9. Ping the PLC and confirm zero packet loss.
10. Use `Get-NetNeighbor` to verify that Windows has learned the PLC's IP-to-MAC mapping.
11. Return to CLICK Programming Software and connect to the discovered PLC.

Result

The final laboratory network uses 192.168.0.20/16 for the Windows USB Ethernet interface and 192.168.0.10/16 for the CLICK PLC. The unmanaged switch provides the Ethernet connection between them; no router, DHCP server, default gateway, or DNS server is required for direct PLC communication.